

# Hu Zhiheng Henrey

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## SUMMARY

Final-year Computer Science (Interactive Media & Game Development) student at Singapore Institute of Technology-DigiPen Institute of Technology joint degree programme. Learns by building, applying C++ and systems programming across multiple projects.

## EDUCATION

### Singapore Institute of Technology (SIT)

Expected graduation 2028

*BSc (Hons) Computer Science in Interactive Media and Game Development [Joint-Degree with DigiPen Institute of Technology Singapore]*

**Certifications:** Google Data Analytics Professional Certificate (Coursera), 2024

## PROJECTS

### 2D platformer where a frog swings, climbs and fights using its stretchy tongue

Spring 2026

*C++, custom engine, 8-person team*

- Built a JSON save and load system with RapidJSON covering about 30 data types, reusable templates and grid data, with categorised crash logging.
- Designed and solely wrote an undo/redo system for the team's in-house design tool, using JSON snapshots to reverse edits, creation and deletion.
- Developed a UI framework with 10 screens, buttons and sliders, supporting keyboard, mouse and gamepad input.
- Implemented behaviour logic as a 4 state machine in C++, after prototyping 4 behaviour types in Unity C#.
- Built an auto-save system that writes progress to a JSON file and restores the last saved state on failure.
- Served as AI & UI Champion on an 8 person team, owning enemy behaviour logic and the UI framework across 6 milestones.

### Stealth maze game where rocks emit waves to reveal the dark and lure enemies away

Spring 2025

*C++, custom engine, 5-person team*

- Designed a state management system where every screen follows the same lifecycle, so screens switch cleanly with no memory leaks.
- Built the full user interface flow across 8 screens and overlays, including menus, settings, pause and results pages.
- Built a reusable UI component library with buttons, mouse hover detection and a font cache shared across all screens.
- Built a draggable volume slider linked to the app's global audio settings.
- Built the rendering system for the map's walls and boundaries.
- Led integration of four teammates' modules into one working application with end conditions.

## EXPERIENCE

### Singapore Civil Defence Force (SCDF) — Emergency Medical Technician

Apr 2022 – Feb 2024

- Delivered emergency medical care as part of an ambulance crew over 22 months, making real-time triage and treatment decisions under pressure. Worked across a rotating team through high-stress emergency callouts, building the resilience to sustain performance under repeated pressure.

### Private Tutor

2020 – Present

- Tutored 4 students in Geography, Social Studies, Maths and Science, raising one student from a fail to B3 in Geography and Social Studies.
- Broke down difficult concepts and adapted lessons to each student's pace and progress, building my communication, patience and empathy.

### TongXin Technology — Industry Attachment

Dec 2019 – Jan 2020

- Observed engineers building an Arduino sensor-guided car and an OpenCV camera pipeline, gaining early exposure to embedded systems and computer vision.

## SKILLS AND INTERESTS

**Technical Skills:** C++, C#, Python, JavaScript, SQL, HTML, Unity, Unreal Engine, Excel, Figma, OpenCV, Arduino, Git

**Interests:** Video/Photo Editing (Affinity, DaVinci Resolve), Dizi (Grade 9), Floorball (SCDF Inter-Division Champion, 2023)